TELECOM / NETWORK ANALYTICS

Reliance Jio Network Experience & Service Reliability Intelligence Dashboard



A Power BI analysis of public TRAI telecom data, 2026

Devika R | Student ID AF05331703

Anudip Foundation | Course DBA_AI | Batch ANP-D7635

Problem statement

Telecom companies must continuously monitor network quality, service reliability, customer support and speed across telecom circles. This project turns public data into an interactive Power BI dashboard.

Compare

Providers and telecom circles on the same indicators

Track

Performance trends over time, month by month

Spot

Areas that may need attention

Project objectives

Network reliability

Availability, call setup, call-drop and packet-drop measures

Customer service

Monitor call setup and customer support performance

Providers and circles

Compare providers and analyse circle-wise performance

Speed and tariff

P80 download and upload trends against tariff information

Data preparation

MySQL and SQL for storage, Power Query for transformation

Dashboards

Power BI and DAX for KPIs, trends and detailed tables

Business questions answered

Availability



What is overall network availability, and how does it change over time?

Call performance



What is call setup success, and how do call and packet drops change by month?

Circles and speed



Which circles perform stronger or weaker, and how does P80 download speed vary?

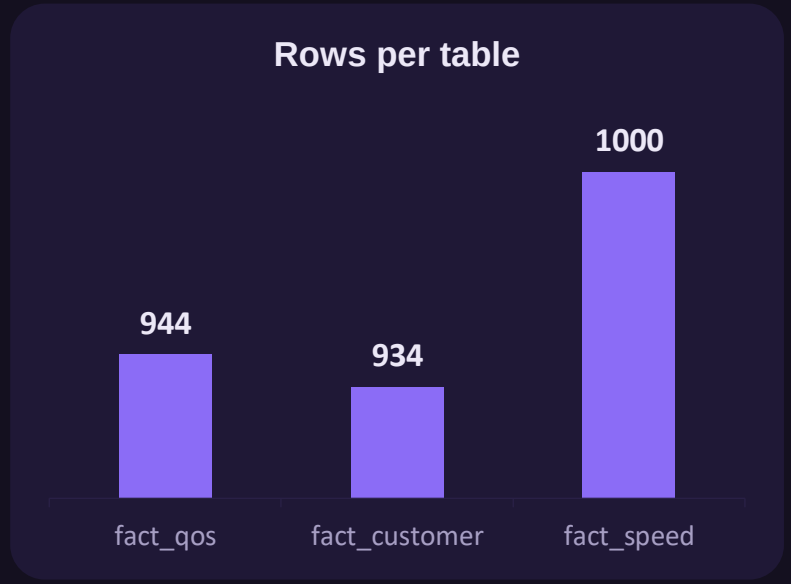
Jio versus market



How does Reliance Jio compare with other providers in the public dataset?

Dataset and tools

Table	Rows	Content
fact_qos	944	Network availability, call and packet drops
fact_customer	934	Customer service indicators
fact_speed	1,000	P80 download/upload speed, tariff category



Source: TRAI public telecom data (Jio Telecom Network Performance Dataset). Columns include provider, circle/LSA, month, availability, call and packet drops, customer service indicators, P80 speeds and tariff category.

Tools: MySQL | SQL | Power Query | Power BI Desktop | DAX | CSV / Excel

Project workflow

1 Industry selection

2 Problem identification

3 Dataset collection

4 MySQL database setup

5 Cleaning with SQL

6 SQL data analysis

7 Power BI modeling

8 Dashboard and insights

Database and SQL analysis

```
CREATE DATABASE project;
USE project;
SELECT * FROM fact_qos;
SELECT * FROM fact_customer;
SELECT * FROM fact_speed;
```

These statements set up the database context and preview the three imported tables. Data cleaning and further analysis are kept in the project SQL files, and Power BI then models the three fact tables for KPIs.



fact_qos



fact_customer



fact_speed

Dashboard page 1: Network Performance Overview

KPI cards

Network availability, call setup success, average latency, packet drop rate

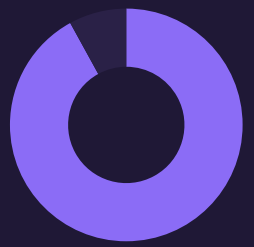
Trends

Availability trend, monthly data call drops, monthly packet drop rate

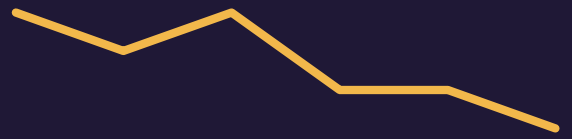
Comparisons

P80 download speed by circle, provider-wise performance, detail table

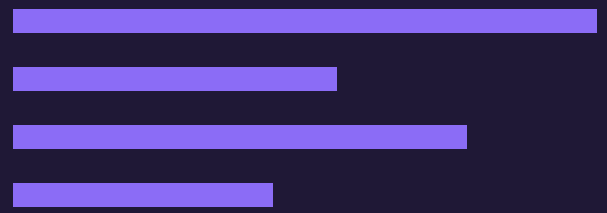
Network availability



Monthly call drops



P80 speed by circle



Charts show the chart style used on this page (illustrative shapes, not actual results)

Dashboard page 2: Customer Service & Reliability

KPI cards

Call center accessibility, calls answered within 90 seconds, bill resolution, bill complaints, closure within 7 days

Ranking

Gauge for 90-second answering and monthly provider ranking

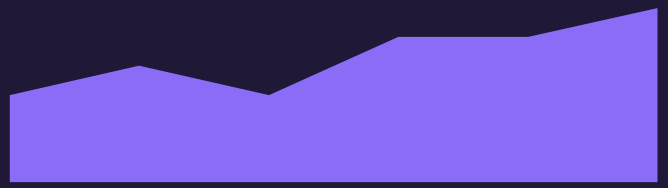
Support view

Support performance by provider and Jio performance by circle

Answered within 90 seconds



Monthly provider ranking



Support by provider



Charts show the chart style used on this page (illustrative shapes, not actual results)

Dashboard page 3: Speed, Circle & Tariff

Monthly trends

P80 download and upload speed month by month

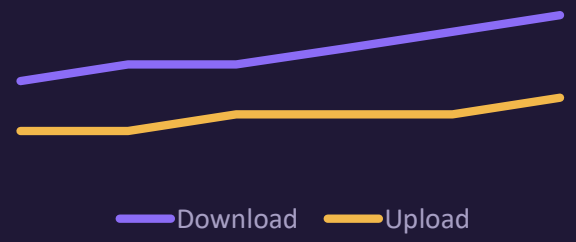
Circle and tariff

Speed comparison across telecom circles and tariff categories

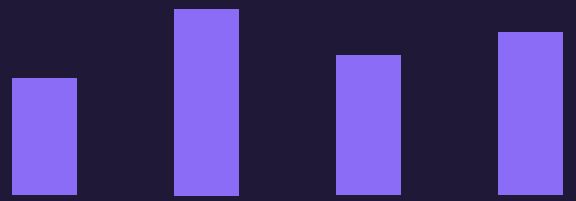
Tariff view

Average P80 download speed by tariff category, plus speed KPI cards

Download vs upload



Speed by circle



Download by tariff



Charts show the chart style used on this page (illustrative shapes, not actual results)

Key insights

Monitoring

One view of network availability and service quality

Trends

Monthly visuals reveal shifts in availability, call drops and packet drops

Providers and circles

Clear differences across providers and service areas

Speed

P80 download and upload vary by month, circle and tariff category

Conclusion and next steps

The dashboard gives one interactive view of network experience, customer service, reliability and speed, built from public data only.

Extend the data



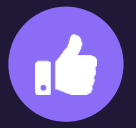
Add more months and quarterly reports

Deepen analysis



More SQL analyses and Jio-versus-market benchmarks

Thank you



Questions welcome

Thank You

Questions and discussion



Devika R | Student ID AF05331703

Anudip Foundation | Course DBA_AI | Batch ANP-D7635